



GAME-SPECIFIC FACULTY GUIDE

Signal House

A two-case economic escape room built around evidence, causal reasoning, and aggregate supply.

Signal House asks students to reconstruct an economic event from records scattered through a house. This guide explains how its two investigations turn those discoveries into economic reasoning and suggests ways to bring the evidence into an AD–AS discussion.

Why This Game Exists

In designing Signal House, I wanted to try a different kind of game. Escape rooms are common, but I had not seen many where the economics itself allowed the player to make progress. That became the idea behind the game: students search the environment, recover records, and gradually work out what must have happened economically.

The format suits a problem economists often face. We see parts of an event before we understand the whole: a household has less left over, a producer faces a higher cost, or a national report shows weaker output. Students have to connect those observations and explain why they belong together.

Some interactions are ordinary escape-room searches. The economic mechanisms then ask students to compare accounts, interpret production records, weigh a business decision, and assemble a causal explanation. Their reasoning releases the next part of the investigation. By the time the formal model appears, they have evidence that gives the curves meaning.

What Students Do

Students explore one continuous house, examining objects and recovering records. The Broken Signal opens a later investigation, The Second Harvest, which reuses the house with new evidence. Students can search broadly, revisit discoveries, and use the satchel to review retained records. Some become useful only when another discovery supplies the missing connection.

A household account may establish what happened to one household; a production record may explain one firm's decisions. Students encounter these fragments before they can make an economy-wide claim. National observations and wider reporting give the local records a broader context. Opening the exit requires students to connect the economic events, whose sequence may differ from the order of discovery.

Both investigations are untimed. Students can use increasingly explicit hints when an object, interaction, or dependency blocks progress, and revise unsuccessful settings or arrangements as they learn more. Hint use carries no score or mastery penalty. This support keeps navigation difficulties from blocking the economics, leaving students to concentrate on connecting the evidence.



One view of the shared house. Students encounter evidence within the environment and bring records together as their investigation develops.

What the Game Is Really Teaching

Evidence and causal explanation. Students build a case from timing, costs, production decisions, and national observations. In *The Broken Signal*, archived reports link storm damage and expensive emergency routes to the production records. A higher price alone leaves several explanations open. Students need a mechanism that accounts for the pattern across records, distinguishing local evidence from support for an economy-wide claim.

Costs, marginal decisions, and scarce resources. The workshop asks students to compare the added revenue and added cost of work that uses limited material. An already-paid, unrecoverable lease is a sunk cost. Material can remain in storage, so exhausting the stock is not the objective. This gives students a concrete reason why a firm may reduce activity even while customer orders remain available.

From firms to short-run aggregate supply. Higher costs across producers reduce the output firms are willing to supply at a given price level. In *The Broken Signal*, production and staffing cuts connect to national evidence of lower real output, higher unemployment, and higher prices. Students interpret this as a leftward shift in short-run aggregate supply (SRAS). Evidence of wider effects supports the move from a firm's experience to an aggregate conclusion.

Demand as a separate condition. Both case interpretations hold aggregate demand (AD) unchanged when identifying the supply shift. In the first, continuing customer orders help students understand the firm's cost-driven response, while the final model explicitly fixes AD. In the second, a separate demand log covers both policy and independent spending conditions. An unchanged policy setting alone would leave other sources of demand change unresolved.

What the measures tell us. National output is measured at constant prices. A household's nominal pay can rise while its purchasing power over an unchanged essential bundle deteriorates. The price index describes a price level; inflation describes its change over an interval. Mission 1 compares matching monthly inflation periods. A change in the price level during adjustment does not establish an indefinitely rising or falling inflation path.

Policy tradeoffs. After the adverse supply shock, tighter demand policy can ease inflation pressure while weakening output and employment further. A looser response can support work and output while adding inflation pressure. Students must recognize both objectives, and neither response repairs the disrupted input network. The policy gauges are illustrative directional indices, so students use them to examine competing objectives without treating the values as calibrated forecasts.

The Second Harvest

The later case begins with a resource-saving trial: three pump-hours in place of six for the same harvest, quality, land, and other inputs. With the operating rate and other batch costs unchanged, the cost of the same batch falls from **\$21 to \$15**. Students reconstruct that cost from separate records.

A successful local trial alone cannot establish an aggregate supply shift. Installation records show that growers, mills, and equipment makers adopted the method across districts and a national supplier network. Students need that wider adoption alongside the cost evidence, national observations, and separate unchanged-demand condition.

The national real-output index rises from **200 to 216**, while the price index falls from **100 to 96**. Together, the records support a rightward SRAS shift with AD unchanged. Establishing that a beneficial method spread gives students a different problem from tracing the earlier disruption.

The comparison establishes a lower price level, without implying that prices keep falling. No corresponding labor-market series is supplied, so employment and unemployment effects remain open.

Comparing the Two Cases

Ask students what changed first in production, what justified extending the explanation to the national economy, and what supports holding AD unchanged. Use the table to compare the evidence behind each shift.

The Broken Signal	The Second Harvest
Disrupted input deliveries and costly emergency routes raise production costs.	A resource-saving method lowers the cost of the same output batch.
Reports across manufacturers and matched national indicators establish wider effects.	Adoption across producer groups and national records establish wider effects.
With AD unchanged, SRAS shifts left: real output falls and the price level rises.	With AD unchanged, SRAS shifts right: real output rises and the price level falls.

The Educational Loop

Mission 1 closes with an evidence recap, economic explanation, labeled AD–AS graph, and transfer forecast. Students can revise their forecast after feedback before opening the results screen. Mission 2 concludes with its evidence-based explanation and comparison of the cases; the explicit graph and transfer forecast belong to Mission 1.

Experience → Consequence → Economic Explanation → Formalization → Transfer

Bringing in the AD–AS Model

1. Ask what students thought was happening before opening the exit and which records changed or confirmed their interpretation.
2. Have them build a causal account, separating local observations from evidence of national change.
3. Draw AD and SRAS with real output on the horizontal axis and the price level on the vertical axis. State the unchanged-AD condition.
4. Map The Broken Signal to SRAS left. Ask students to justify the output and price-level changes using their records.
5. Map The Second Harvest to SRAS right. Ask which evidence supports lower costs and wider adoption.
6. Compare the evidence supporting each interpretation, including unchanged demand, and ask what would make students reconsider.

Using It in Class

These are optional ways to use the investigation; choose what fits your course.

Before formal AD–AS instruction

Let students investigate and explain what the records suggest in everyday language. They can reconstruct the mechanism before knowing its formal name. After play, use their evidence to introduce the graph.

During AD–AS instruction

Use The Broken Signal after introducing supply-side shifts. Students can pause when the first door opens to construct the graph together before its reveal. The Second Harvest then supplies a contrasting case.

After instruction

Use either case, subject to the second mission's unlock requirement, as evidence-based practice. Ask students to defend the supply shift with records from the investigation and explain what rules out a competing interpretation.

Independent/student review

Students can revisit evidence and work through Mission 1's graph and forecast. Mission 2 unlocks at the first exit, independently of the forecast, so invite students to complete that follow-up. You can request a reflection or screenshot through your usual course workflow; the game sends no faculty completion report.

Questions to Consider

Select any of these prompts for discussion or a short reflection.

1. Which record first made you suspect a production-side change? What other explanation was still possible at that point?
2. What additional evidence did you need before treating a local cost or production change as economy-wide? Compare the two cases.
3. In *The Broken Signal*, how did the cost records explain firms' decisions while customer orders remained available? Which costs mattered to accepting additional work?
4. Why did the first case create a difficult policy choice? What could the policy control change, and what remained unresolved?
5. In *The Second Harvest*, what did holding harvest, quality, and other inputs constant let you infer from fewer pump-hours?
6. Why was the local trial insufficient for an aggregate conclusion? Which evidence established that other producers had adopted the improvement?
7. What supports holding AD unchanged in each case? Distinguish evidence about one firm's orders from the broader demand condition used in the model.
8. A third case reports rising output and a rising price level. What additional evidence would you request before deciding which curve shifted?

Common Misconceptions

- **“Higher prices mean demand increased.”** Ask students to explain the cost and output evidence alongside the price change.
- **“One firm's cost change proves aggregate supply shifted.”** Look for evidence of wider exposure or adoption and national consequences.
- **“The rise in unemployment caused the supply shock.”** Revisit the timing: the first case links the input disruption to costs, production cuts, and weaker employment.
- **“The policy machine has one setting that fixes everything.”** Ask students to describe what happened to both objectives in each trial and whether the input network was repaired.
- **“A lower price level means continuing deflation.”** A comparison between two price levels establishes a change over that interval; its continuation needs further evidence.
- **“A productive local trial proves economy-wide growth.”** Separate the trial result from adoption and national output evidence. The second case establishes a short-run supply improvement, without measuring a permanent growth rate.